Lab HA/BCP — Building a Proxmox cluster and testing it under failure

GRETA de l'Essonne — AIS professional title — Block 2: Design and evolution

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Charles Maduba — Group 3 (work carried out in a team of 4 trainees)

Session A: Monday 5 October 2026 — Parts 1 and 2. Session B: Thursday 8 October 2026 — Parts 3 and 4.

English version of the lab report, for personal reference. The original report was written in French.

Summary

Part	Content	Status
1	Prepare 3 Proxmox servers (pve1, pve2, pve3)	Done
2	Create the cluster and check quorum	Done
3	Shared storage (NFS)	Done
4	HA, failure test, RTO measurement	Done (RTO: 2 min 17 s)

Node	Name	IP address
Node 1 (existing)	pve (alias pve1)	192.168.56.200
Node 2	pve2	192.168.56.201
Node 3	pve3	192.168.56.202

PART 1 — Preparing the 3 servers

1.1 — Creating pve2 and pve3

Note: pve2 and pve3 were not cloned from pve1. They were installed from scratch from the Proxmox VE ISO, in VirtualBox, on the same host-only network 192.168.56.0/24. There are therefore no screenshots of the clone wizard.

Consequence: each node got its own SSH keys and its own machine identifier (machine-id) at installation. Step 1.3 (regenerating keys and machine-id) was not needed; we only checked that the three identifiers are different.

1.2 — Name, /etc/hosts file and IP address of each node

```
root@pve:~# cat /etc/hosts
127.0.0.1 localhost.localdomain localhost
192.168.56.200 pve.local.lan pve pve1
192.168.56.201 pve2.lab.local pve2
192.168.56.202 pve3.lab.local pve3

# The following lines are desirable for IPv6 capable hosts

::1     ip6-localhost ip6-loopback
fe00::0 ip6-localnet
ff00::0 ip6-mcastprefix
ff02::1 ip6-allnodes
ff02::2 ip6-allrouters
ff02::3 ip6-allhosts
root@pve:~# hostname -i
192.168.56.200
root@pve:~# cat /etc/machine-id
f4679ee0fbab48b2bebf6f1afdb5e526
root@pve:~#
```

pve1: /etc/hosts with the 3 nodes, hostname -i = 192.168.56.200, machine-id

```
root@pve2:~# cat /etc/hosts
127.0.0.1 localhost.localdomain localhost
192.168.56.201 pve2.lab.local pve2
192.168.56.200 pve.local.lan pve pve1
192.168.56.202 pve3.lab.local pve3

# The following lines are desirable for IPv6 capable hosts

::1     ip6-localhost ip6-loopback
fe00::0 ip6-localnet
ff00::0 ip6-mcastprefix
ff02::1 ip6-allnodes
ff02::2 ip6-allrouters
ff02::3 ip6-allhosts
root@pve2:~# hostname -i
192.168.56.201
root@pve2:~# ip -br a
lo               UNKNOWN        127.0.0.1/8 ::1/128
enp0s3           UP
enp0s8           UP             10.0.3.15/24 fd17:625c:f037:3:a00:27ff:feef:ef90/64 fe80::a00:27ff:feef:ef90/64
vmbr0            UP             192.168.56.201/24 fe80::a00:27ff:fef6:a204/64
root@pve2:~#
```

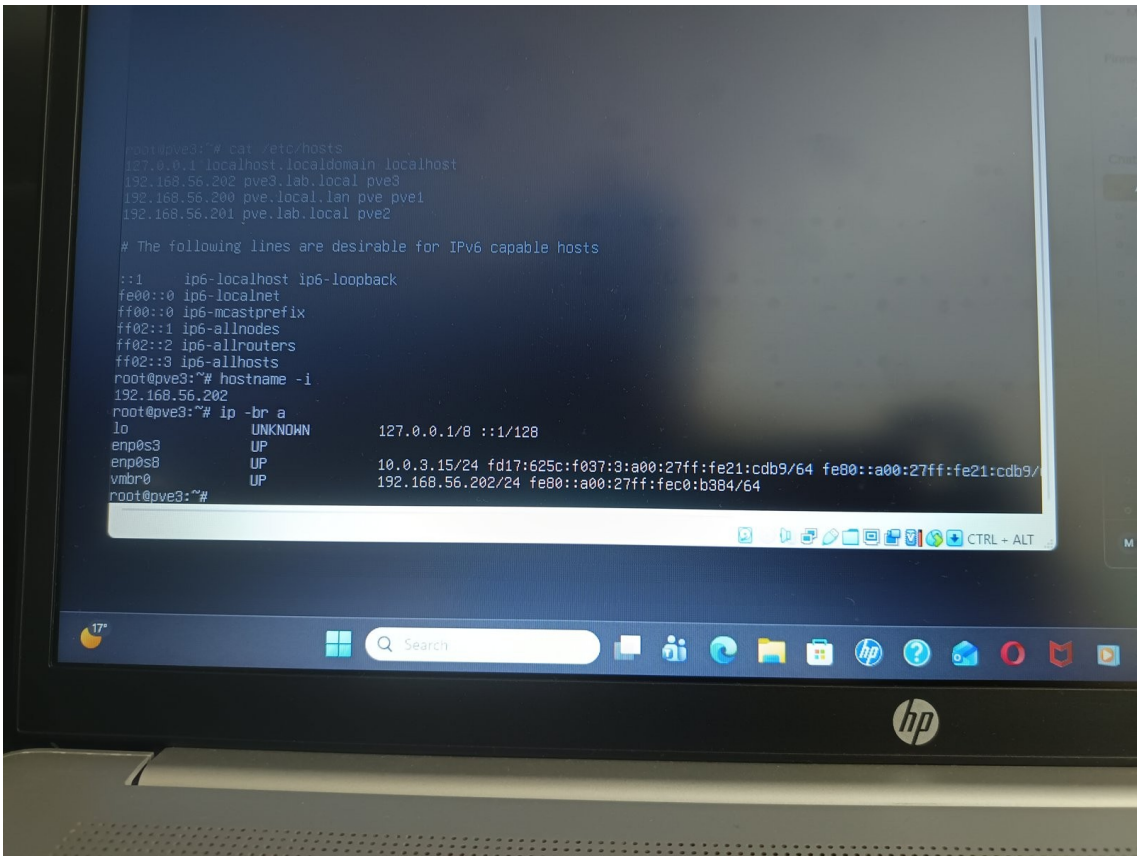
pve2: /etc/hosts, hostname -i and ip -br a (vmbr0 = 192.168.56.201)

```
root@pve3:~# sed -i 's/201 pve.lab.local/201 pve2.lab.local/' /etc/hosts
root@pve3:~# cat /etc/hosts
127.0.0.1 localhost.localdomain localhost
192.168.56.202 pve3.lab.local pve3
192.168.56.200 pve.local.lan pve pve1
192.168.56.201 pve2.lab.local pve2

# The following lines are desirable for IPv6 capable hosts

::1     ip6-localhost ip6-loopback
fe00::0 ip6-localnet
ff00::0 ip6-mcastprefix
ff02::1 ip6-allnodes
ff02::2 ip6-allrouters
ff02::3 ip6-allhosts
root@pve3:~#
```

pve3: contents of /etc/hosts



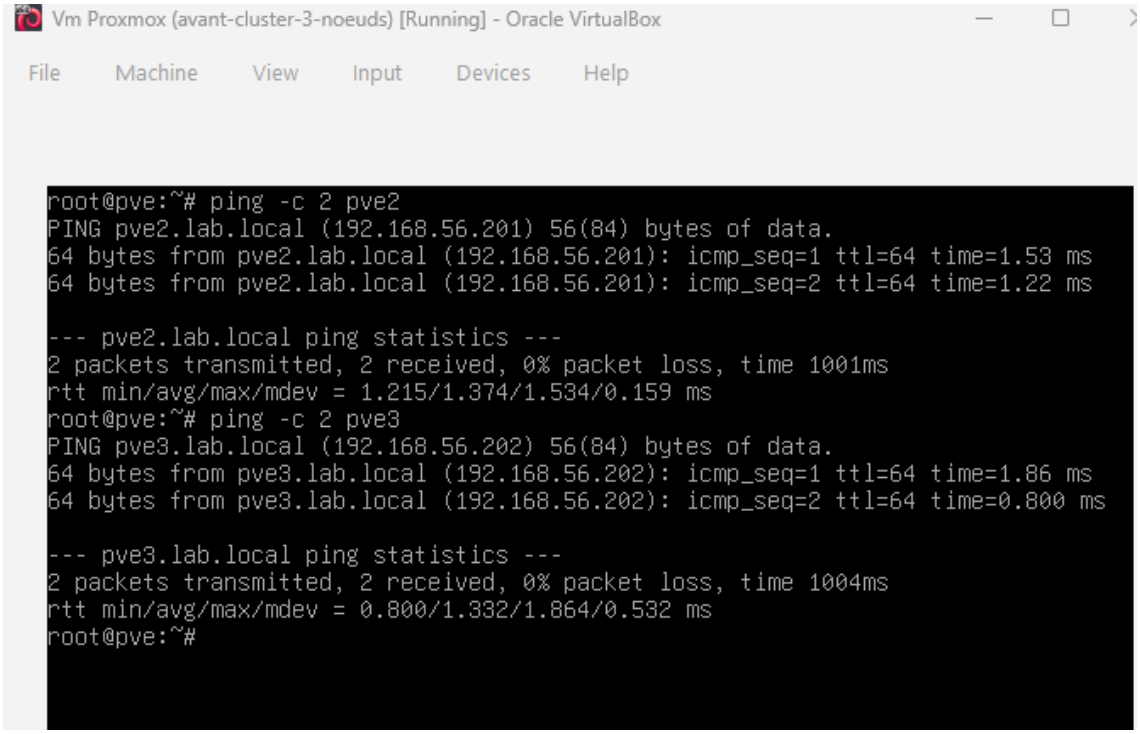
pve3: hostname -i and ip -br a (vmbr0 = 192.168.56.202)

1.3 — The 3 machine-ids

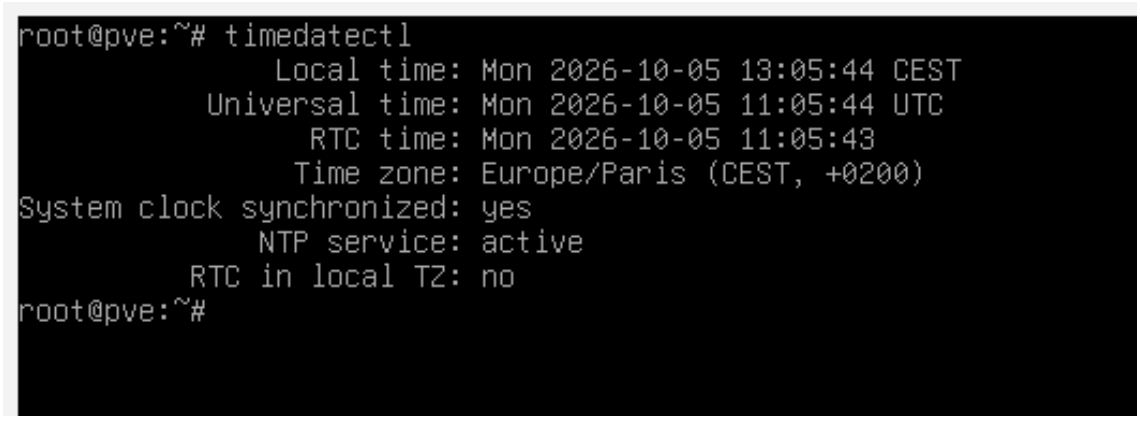
Node	machine-id
pve1	f4679ee0fbab48b2bebf6f1afdb5e526
pve2	0c5c5b5745d04cf2985687e8c53da75f
pve3	c895ad2836d5419fb7c48d0328d078e3

The three identifiers are different (screenshots above for pve1, below for pve2 and pve3).

1.4 — The 3 nodes can see each other (ping, time, empty nodes)



pve1: ping -c 2 pve2 and ping -c 2 pve3 — each name answers with the right IP address



pve1: timedatectl — System clock synchronized: yes

```
root@pve2:~# cat /etc/machine-id
0c5c5b5745d04cf2985687e8c53da75f
root@pve2:~# ping -c 2 pve1
PING pve.local.lan (192.168.56.200) 56(84) bytes of data.
64 bytes from pve.local.lan (192.168.56.200): icmp_seq=1 ttl=64 time=1.33 ms
64 bytes from pve.local.lan (192.168.56.200): icmp_seq=2 ttl=64 time=0.620 ms

--- pve.local.lan ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1003ms
rtt min/avg/max/mdev = 0.620/0.974/1.329/0.354 ms
root@pve2:~# ping -c 2 pve3
PING pve3.lab.local (192.168.56.202) 56(84) bytes of data.
64 bytes from pve3.lab.local (192.168.56.202): icmp_seq=1 ttl=64 time=5.83 ms
64 bytes from pve3.lab.local (192.168.56.202): icmp_seq=2 ttl=64 time=0.584 ms

--- pve3.lab.local ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1107ms
rtt min/avg/max/mdev = 0.584/3.207/5.831/2.623 ms
root@pve2:~# timedatectl
               Local time: Mon 2026-10-05 12:42:18 CEST
           Universal time: Mon 2026-10-05 10:42:18 UTC
                 RTC time: Mon 2026-10-05 10:42:18
                Time zone: Europe/Paris (CEST, +0200)
System clock synchronized: yes
              NTP service: active
          RTC in local TZ: no
root@pve2:~# qm list
root@pve2:~# pct list
root@pve2:~#
```

pve2: machine-id, ping pve1 and pve3, timedatectl (synchronised), qm list and pct list empty

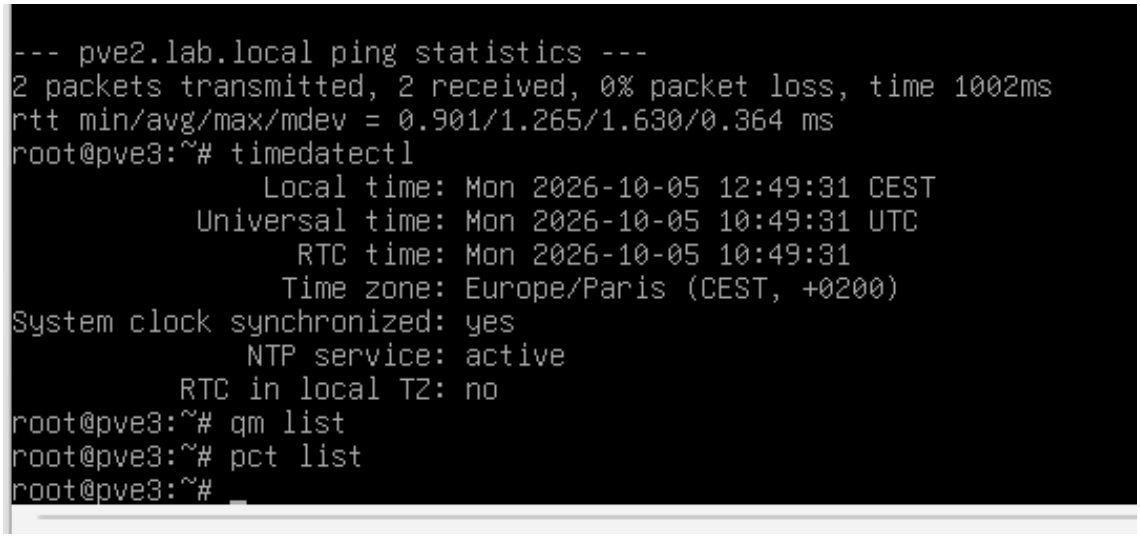
```
192.168.56.202
root@pve3:~# ip -br a
lo               UNKNOWN        127.0.0.1/8 ::1/128
enp0s3           UP
enp0s8           UP             10.0.3.15/24 fd17:625c:f037:3:a00:27ff:fe21:cdb9/64 fe80::a00:27ff:fe21:cdb9
vmbr0            UP             192.168.56.202/24 fe80::a00:27ff:fec0:b384/64
root@pve3:~# sed -i 's/201 pve.lab.local/201 pve2.lab.local/' /etc/hosts
root@pve3:~# cat /etc/hosts
127.0.0.1 localhost.localdomain localhost
192.168.56.202 pve3.lab.local pve3
192.168.56.200 pve.local.lan pve pve1
192.168.56.201 pve2.lab.local pve2

# The following lines are desirable for IPv6 capable hosts

::1     ip6-localhost ip6-loopback
fe00::0 ip6-localnet
ff00::0 ip6-mcastprefix
ff02::1 ip6-allnodes
ff02::2 ip6-allrouters
ff02::3 ip6-allhosts
root@pve3:~# cat /etc/machine-id
c895ad2836d5419fb7c48d0328d078e3
root@pve3:~# ping -c 2 pve1
PING pve.local.lan (192.168.56.200) 56(84) bytes of data.
64 bytes from pve.local.lan (192.168.56.200): icmp_seq=1 ttl=64 time=2.37 ms
64 bytes from pve.local.lan (192.168.56.200): icmp_seq=2 ttl=64 time=0.647 ms

--- pve.local.lan ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1002ms
rtt min/avg/max/mdev = 0.647/1.508/2.370/0.861 ms
root@pve3:~# ping -c 2 pve2
PING pve2.lab.local (192.168.56.201) 56(84) bytes of data.
64 bytes from pve2.lab.local (192.168.56.201): icmp_seq=1 ttl=64 time=1.63 ms
64 bytes from pve2.lab.local (192.168.56.201): icmp_seq=2 ttl=64 time=0.901 ms
```

pve3: ip -br a, /etc/hosts, machine-id, ping pve1 and pve2

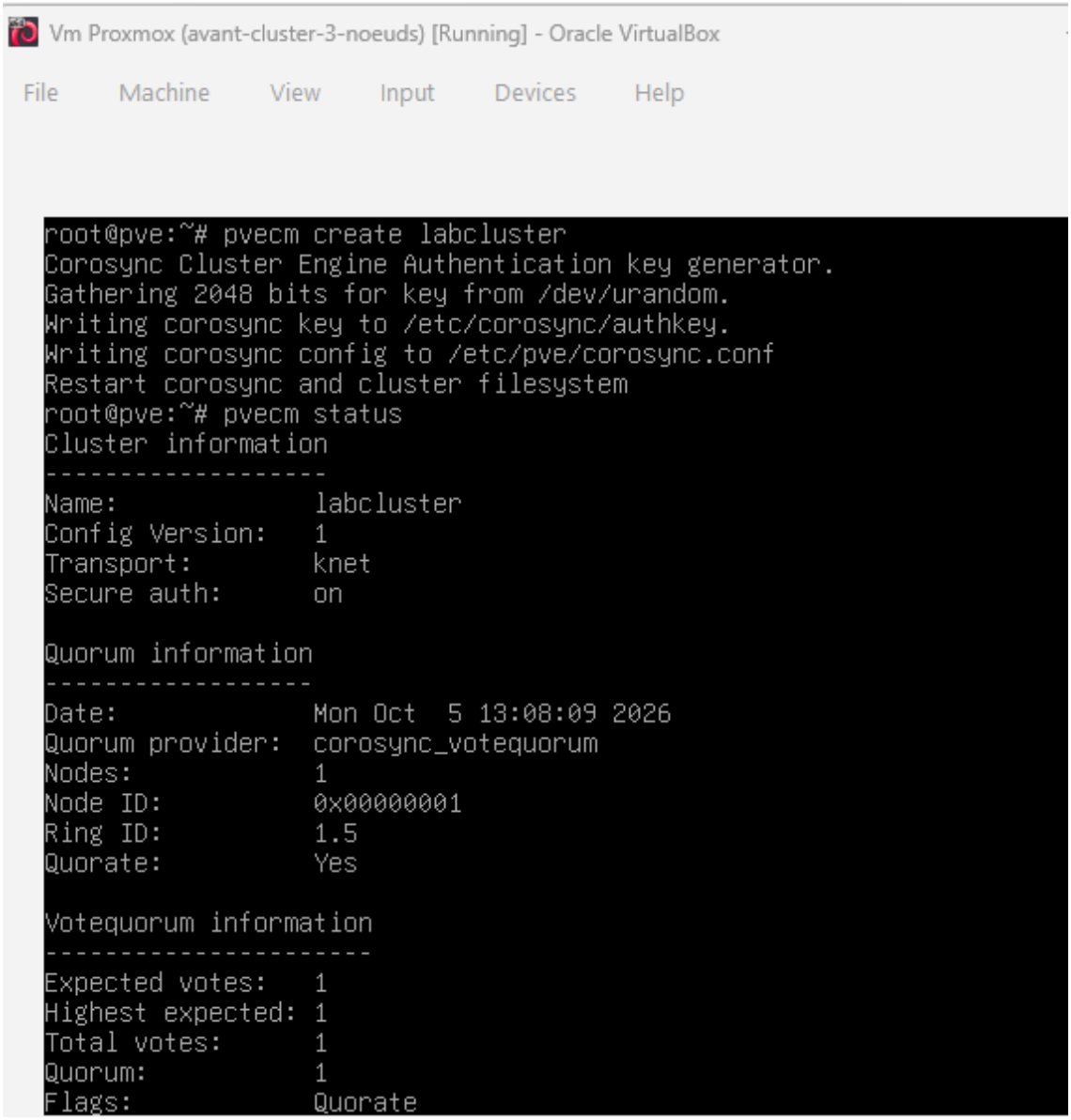


pve3: timedatectl (synchronised), qm list and pct list empty

Result: each name answers with the right IP address, the clock is synchronised on all 3 nodes, and pve2 and pve3 contain no VMs or containers (a requirement to join a cluster).

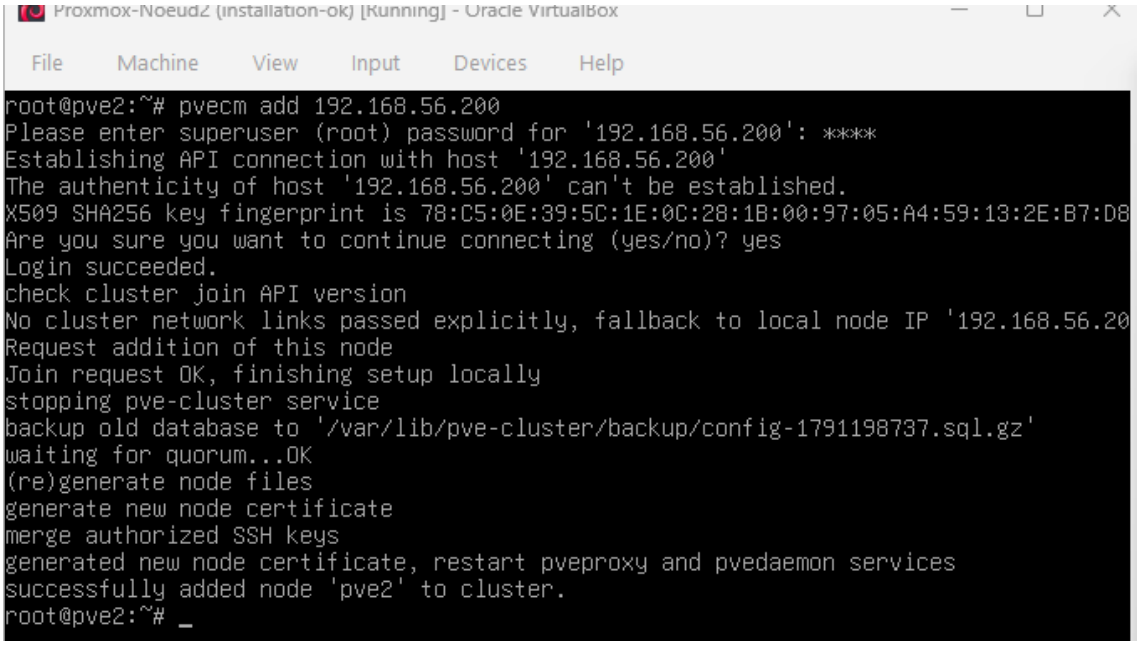
PART 2 — Creating the cluster

2.1 / 2.2 — Creating the cluster on pve1 and first pvecm status



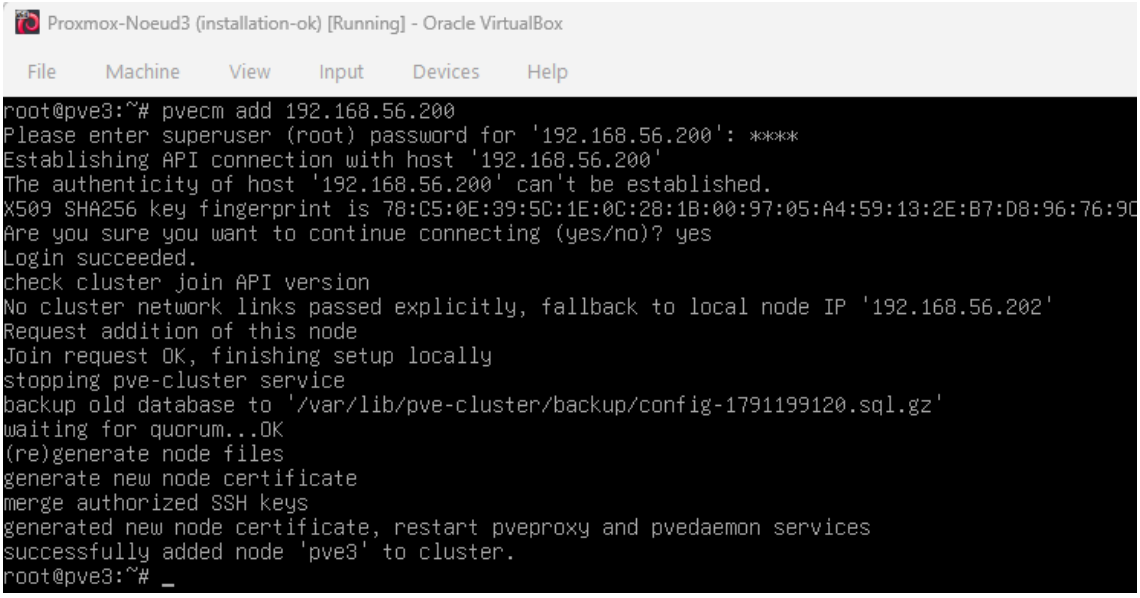
pvecm create labcluster then pvecm status: Nodes: 1, Quorate: Yes

2.3 — Adding pve2



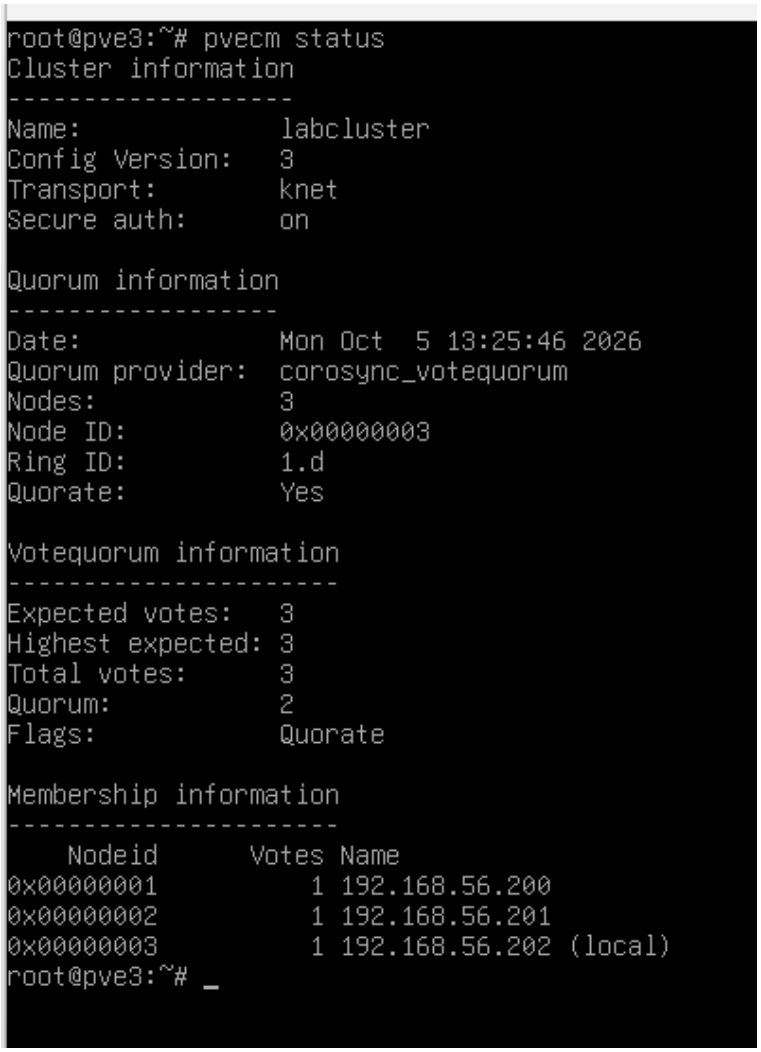
pve2: pvecm add 192.168.56.200 — node added to the cluster

2.4 — Adding pve3



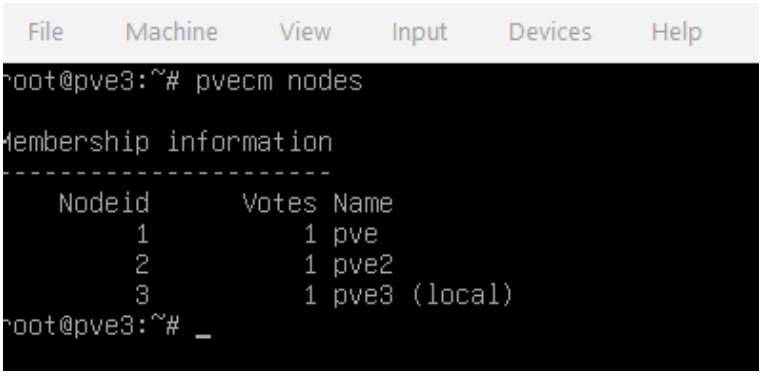
pve3: pvecm add 192.168.56.200 — node added to the cluster

2.5 — pvecm status with 3 nodes



Nodes: 3, Total votes: 3, Quorum: 2, Quorate: Yes

2.6 — pvecm nodes

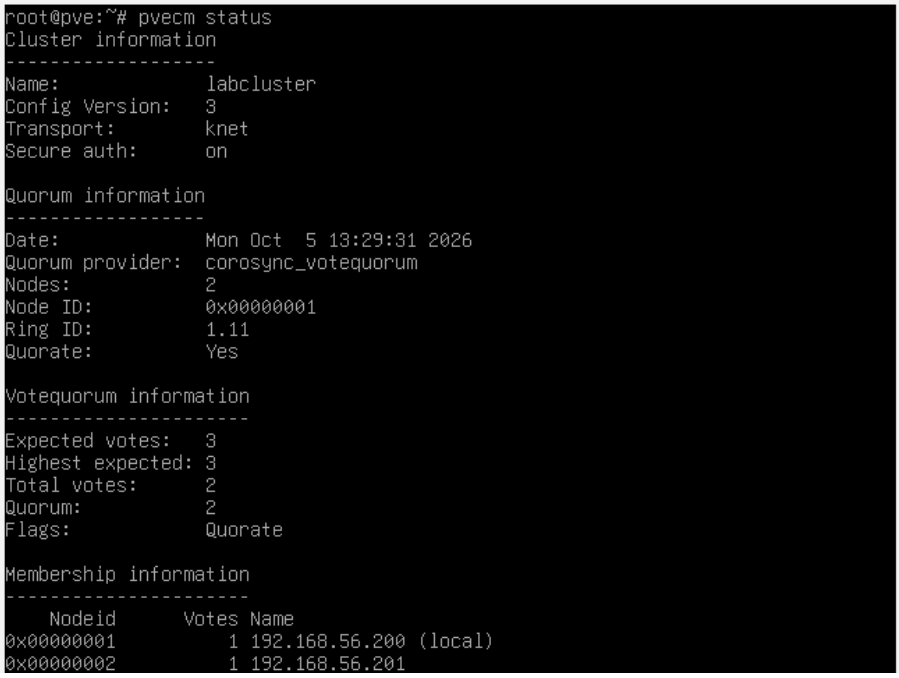


pve, pve2 and pve3 listed

2.7 — Experiment: quorum live

Situation	Votes present	Quorate	Observation

3 nodes on	3 / 3	Yes	Normal cluster
pve3 off	2 / 3	Yes	The cluster keeps working
pve2 and pve3 off	1 / 3	No	"Activity blocked": the cluster refuses to act
pve2 and pve3 back on	3 / 3	Yes	Back to normal after 1 to 2 minutes



pve3 off: Nodes: 2, Quorate: Yes

```
root@pve:~# pvecm status
Cluster information
-------------------
Name:             labcluster
Config Version:   3
Transport:        knet
Secure auth:      on

Quorum information
------------------
Date:             Mon Oct  5 13:32:30 2026
Quorum provider:  corosync_votequorum
Nodes:            1
Node ID:          0x00000001
Ring ID:          1.15
Quorate:          No

Votequorum information
----------------------
Expected votes:   3
Highest expected: 3
Total votes:      1
Quorum:           2 Activity blocked
Flags:

Membership information
----------------------
    Nodeid      Votes Name
0x00000001          1 192.168.56.200 (local)
root@pve:~# _
```

pve2 and pve3 off: Nodes: 1, Quorate: No, "Activity blocked"

```
root@pve:~# pvecm status
Cluster information
-------------------
Name:             labcluster
Config Version:   3
Transport:        knet
Secure auth:      on

Quorum information
------------------
Date:             Mon Oct  5 13:44:12 2026
Quorum provider:  corosync_votequorum
Nodes:            3
Node ID:          0x00000001
Ring ID:          1.1e
Quorate:          Yes

Votequorum information
----------------------
Expected votes:   3
Highest expected: 3
Total votes:      3
Quorum:           2
Flags:            Quorate

Membership information
----------------------
    Nodeid      Votes Name
0x00000001          1 192.168.56.200 (local)
0x00000002          1 192.168.56.201
```

pve2 and pve3 back on: Nodes: 3, Quorate: Yes

Question: why are 2 nodes out of 3 enough, while 1 node out of 3 is not?

Quorum is the majority of votes: quorum = (number of nodes ÷ 2) rounded down + 1. With 3 nodes: 3 ÷ 2 = 1.5 → 1, then 1 + 1 = 2 votes.

With 2 nodes on, there are 2 votes ≥ 2: the cluster has quorum and keeps working.

With only 1 node, there is 1 vote < 2: no quorum, the cluster blocks actions ("Activity blocked"). This is the protection against split-brain: an isolated node must not act alone, because it cannot know whether the others are still working on their side. If it did, two parts of the cluster could start the same VM at the same time and damage its data.

PART 3 — Shared storage (NFS)

Done on the command line (pvesm / qm) instead of the web interface: with the 3 nodes running, the host PC did not have enough memory to open a browser. The result is identical; the web interface uses the same commands.

3.1 — Installing the NFS server on pve and creating the shared folder

Commands: apt install -y nfs-kernel-server then mkdir -p /srv/nfs/vmstore. Before installing, we checked pve's internet access (ping deb.debian.org: 0% loss).

```
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following NEW packages will be installed:
  nfs-kernel-server
0 upgraded, 1 newly installed, 0 to remove and 149 not upgraded.
Need to get 152 kB of archives.
After this operation, 626 kB of additional disk space will be used.
Get:1 http://ftp.fr.debian.org/debian bookworm/main amd64 nfs-kernel-server amd6
4 1:2.6.2-4+deb12u1 [152 kB]
Fetched 152 kB in 0s (1,569 kB/s)
Selecting previously unselected package nfs-kernel-server.
(Reading database ... 46310 files and directories currently installed.)
Preparing to unpack .../nfs-kernel-server_1%3a2.6.2-4+deb12u1_amd64.deb ...
Unpacking nfs-kernel-server (1:2.6.2-4+deb12u1) ...
Setting up nfs-kernel-server (1:2.6.2-4+deb12u1) ...
Created symlink /etc/systemd/system/nfs-client.target.wants/nfs-blkmap.service →
 /lib/systemd/system/nfs-blkmap.service.
Created symlink /etc/systemd/system/multi-user.target.wants/nfs-server.service →
 /lib/systemd/system/nfs-server.service.
nfs-mountd.service is a disabled or a static unit, not starting it.
nfsdcld.service is a disabled or a static unit, not starting it.

Creating config file /etc/exports with new version

Creating config file /etc/default/nfs-kernel-server with new version
Processing triggers for man-db (2.11.2-2) ...
root@pve:~# mkdir -p /srv/nfs/vmstore
root@pve:~#
```

nfs-kernel-server installed without errors, /srv/nfs/vmstore created

3.2 — Declaring the share (exportfs -v)

Line added to /etc/exports: /srv/nfs/vmstore 192.168.56.0/24(rw,sync,no_subtree_check,no_root_squash), then exportfs -ra. Only the lab network (192.168.56.0/24) can access the share.

```
root@pve:~# mkdir -p /srv/nfs/vmstore
root@pve:~# echo '/srv/nfs/vmstore 192.168.56.0/24(rw,sync,no_subtree_check,no_r
oot_squash)' >> /etc/exports
root@pve:~# cat /etc/exports
# /etc/exports: the access control list for filesystems which may be exported
#               to NFS clients.  See exports(5).
#
# Example for NFSv2 and NFSv3:
# /srv/homes       hostname1(rw,sync,no_subtree_check) hostname2(ro,sync,no_subt
ree_check)
#
# Example for NFSv4:
# /srv/nfs4        gss/krb5i(rw,sync,fsid=0,crossmnt,no_subtree_check)
# /srv/nfs4/homes  gss/krb5i(rw,sync,no_subtree_check)
#
/srv/nfs/vmstore 192.168.56.0/24(rw,sync,no_subtree_check,no_root_squash)
root@pve:~# exportfs-ra
-bash: exportfs-ra: command not found
root@pve:~# exportfs -ra
root@pve:~# exportfs -v
/srv/nfs/vmstore
                192.168.56.0/24(sync,wdelay,hide,no_subtree_check,sec=sys,rw,sec
ure,no_root_squash,no_all_squash)
root@pve:~#
```

cat /etc/exports and exportfs -v: the /srv/nfs/vmstore share is active

3.3 — nfs-shared storage added to the cluster (visible on the 3 nodes)

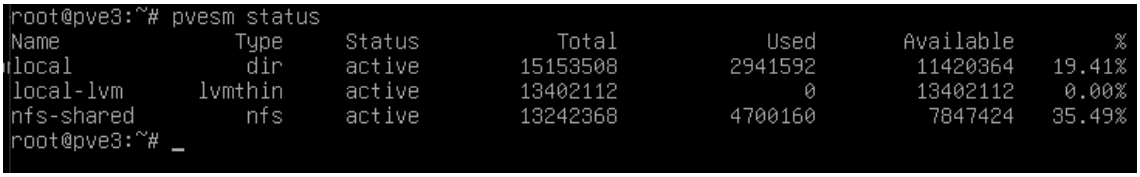
Command on pve: pvesm add nfs nfs-shared --server 192.168.56.200 --export /srv/nfs/vmstore --content images. The storage is defined at Datacenter level: pve2 and pve3 receive it automatically.

```
root@pve:~# systemctl restart pve-cluster
root@pve:~# pvesm add nfs nfs-shared --server 192.168.56.200 --export /srv/nfs/v
mstore --content images
root@pve:~# pvesm status
Name             Type     Status           Total            Used       Availabl
e        %
local             dir     active        13242248         4698152         784962
0   35.48%
local-lvm     lvmthin     active        11460608         9796527         166408
0   85.48%
nfs-shared        nfs     active        13242368         4698112         784998
4   35.48%
root@pve:~#
```

pve: nfs-shared, type nfs, status active

```
root@pve2:~# pvesm status
Name             Type     Status           Total            Used       Available        %
local             dir     active        15153508         2950036        11411920   19.47%
local-lvm     lvmthin     active        13402112               0        13402112    0.00%
nfs-shared        nfs     active        13242368         4700160         7847424   35.49%
root@pve2:~#
```

pve2: nfs-shared active

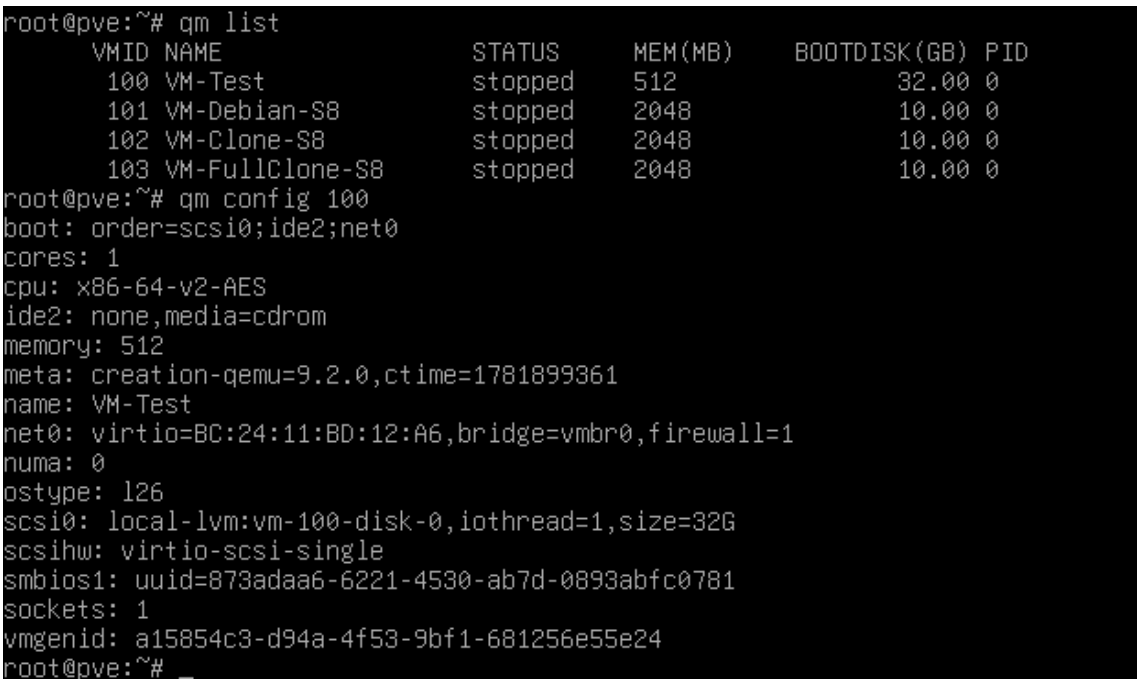


pve3: nfs-shared active

Note: a first attempt failed with "cfs-lock … error: no quorum" while pvecm status showed Quorate: Yes. The cluster configuration service (pmxcfs) had stayed read-only after the nodes briefly lost contact. Fixed with systemctl restart pve-cluster.

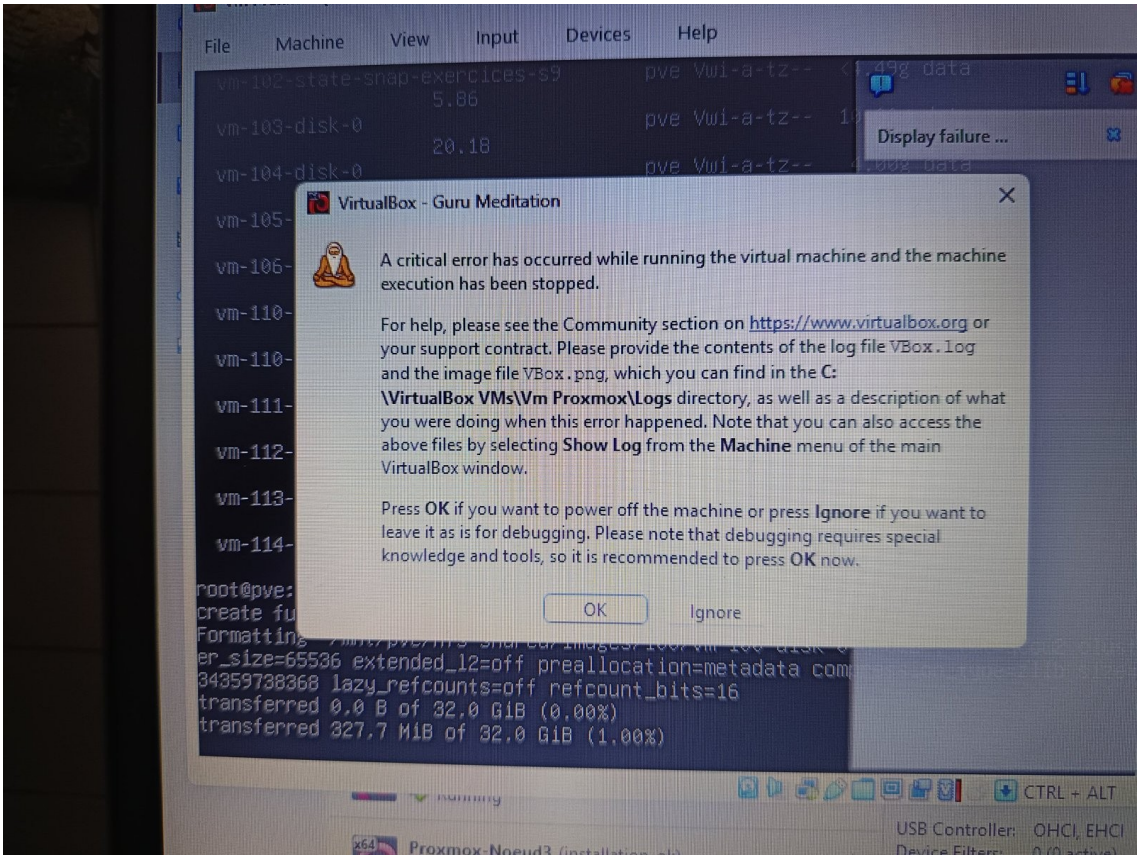
3.4 — VM-Test disk placed on nfs-shared

VM-Test is VM 100 (512 MB RAM, no ISO in the CD drive). Its 32 GB disk was on local-lvm.



qm list and qm config 100 before the operation: disk on local-lvm, 32 GB

Incident: moving the disk (qm disk move, equivalent to "Move Storage") crashed the pve VM (VirtualBox "Guru Meditation") at 1%, because the host PC ran out of resources. The original disk was untouched (deletion only happens after a successful copy). After restarting pve, we removed the incomplete copy with pvesm free.

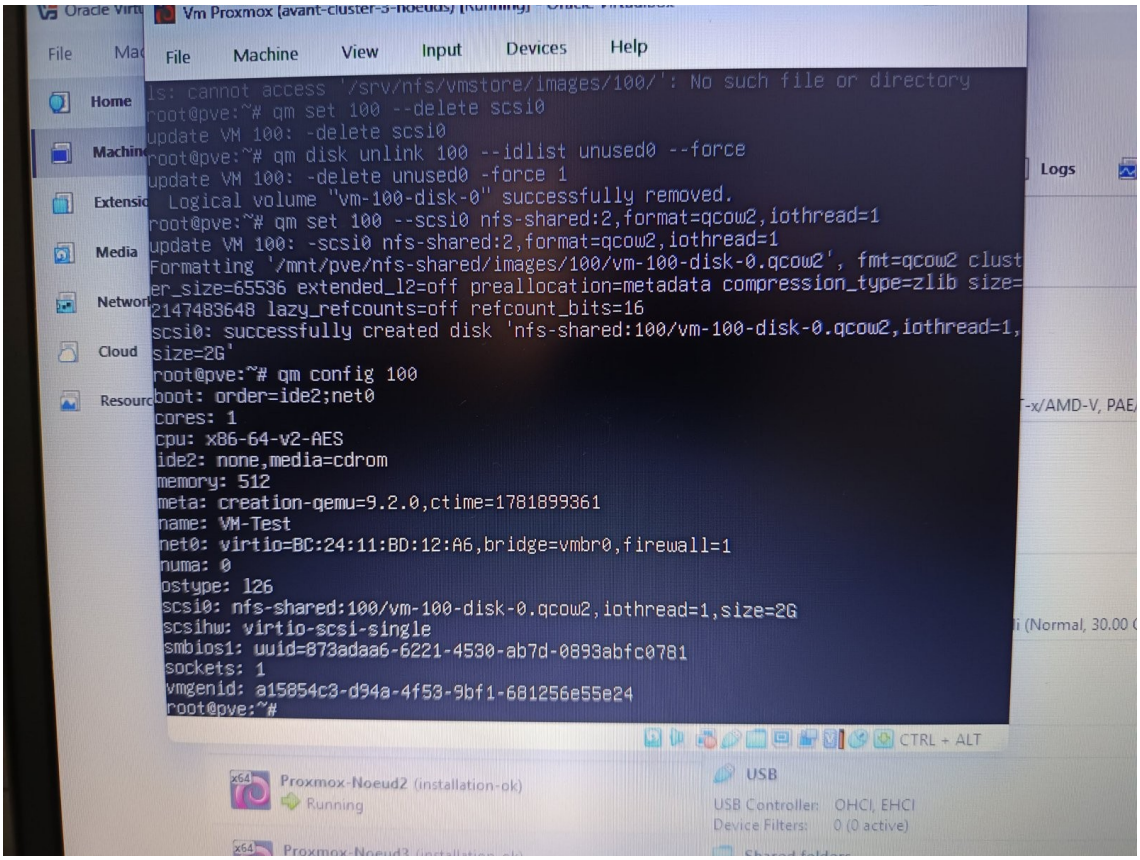


Crash of the pve VM during the disk copy (Guru Meditation)

The lvs command showed that VM-Test's disk was empty (Data% = 0.00). We therefore replaced it with a 2 GB disk created directly on nfs-shared: qm set 100 --delete scsi0, qm disk unlink 100 --idlist unused0 --force, then qm set 100 --scsi0 nfs-shared:2,format=qcow2,iothread=1. Same result for the rest of the lab: the VM's disk is on the shared storage.

```
  vm-100-disk-0                          pve Vwi-a-tz--  32.00g data
                   0.00
  vm-102-disk-0                          pve Vwi-a-tz--  10.00g data snap_vm-102-d
isk-0_snap-apache-ok 20.44
  vm-102-state-snap-apache-ok            pve Vwi-a-tz--  <4.49g data
                   11.80
  vm-102-state-snap-exercices-s9         pve Vwi-a-tz--  <4.49g data
                   5.86
  vm-103-disk-0                          pve Vwi-a-tz--  10.00g data
                   20.18
  vm-104-disk-0                          pve Vwi-a-tz--   4.00g data
                   3.63
  vm-105-disk-0                          pve Vwi-a-tz--   8.00g data
                   2.60
  vm-106-disk-0                          pve Vwi-a-tz--   8.00g data
                   2.60
  vm-110-disk-0                          pve Vwi-a-tz--   2.00g data
                   19.69
  vm-110-disk-1                          pve Vwi-a-tz--   2.00g data
                   32.53
  vm-111-disk-0                          pve Vwi-a-tz--   2.00g data
                   29.52
  vm-112-disk-0                          pve Vwi-a-tz--   2.00g data
                   32.57
  vm-113-disk-0                          pve Vwi-a-tz--   2.00g data
                   32.58
  vm-114-disk-0                          pve Vwi-a-tz--   2.00g data
                   32.53
root@pve:~# _
```

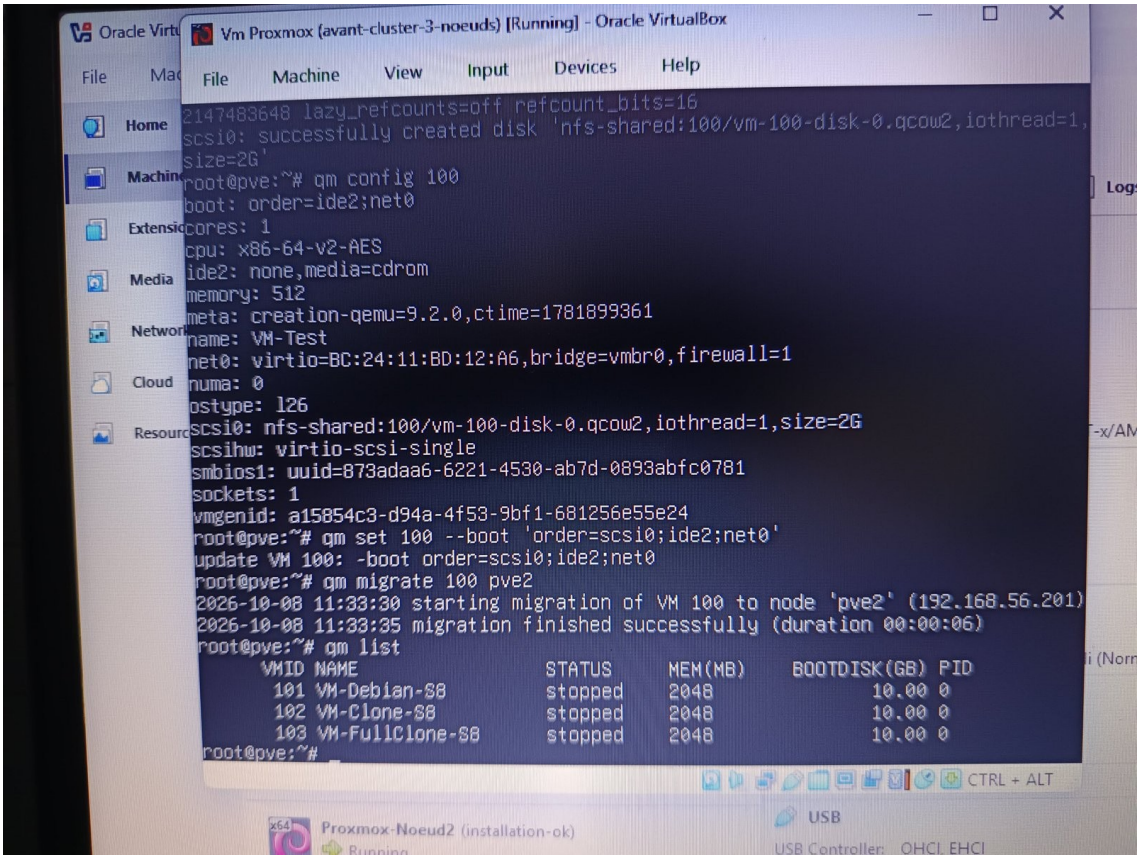
lvs: vm-100-disk-0 at 0.00% (empty disk)



qm config 100: scsi0: nfs-shared:100/vm-100-disk-0.qcow2, size=2G

3.5 — Migrating VM-Test to pve2

Boot order restored (qm set 100 --boot 'order=scsi0;ide2;net0'), then qm migrate 100 pve2. The migration took 6 seconds: only the VM's configuration moves; the disk stays on nfs-shared and is not copied.

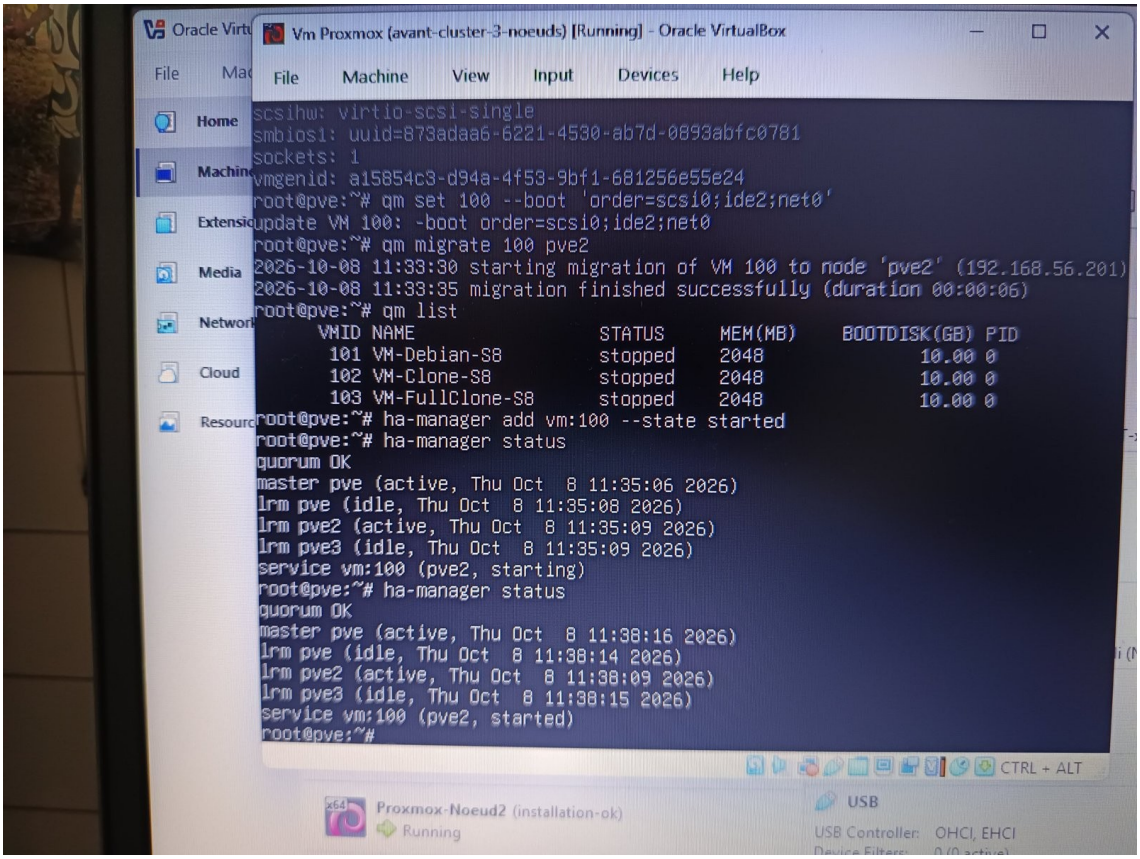


migration finished successfully (duration 00:00:06); VM 100 no longer listed on pve

PART 4 — High availability and failure test

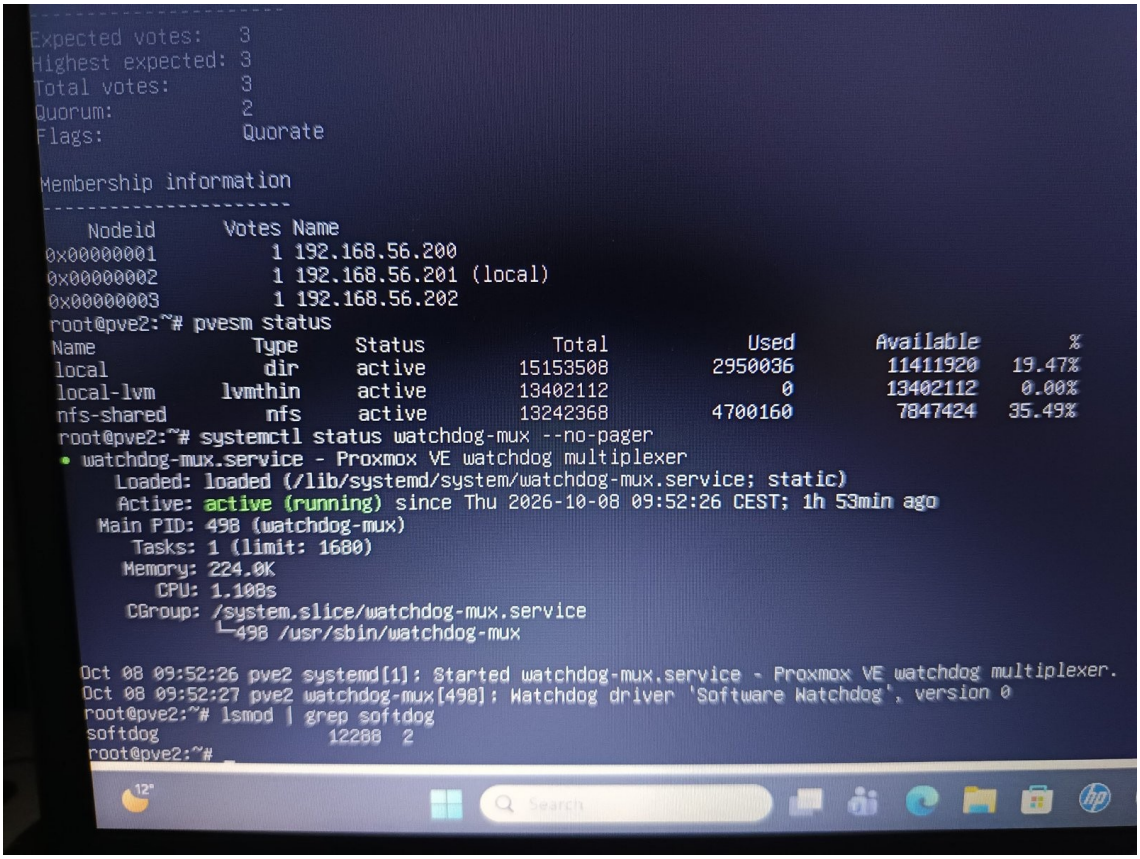
4.1 — Enabling HA on VM-Test

Command: ha-manager add vm:100 --state started. HA starts the VM on pve2.



ha-manager status: quorum OK, service vm:100 (pve2, started)

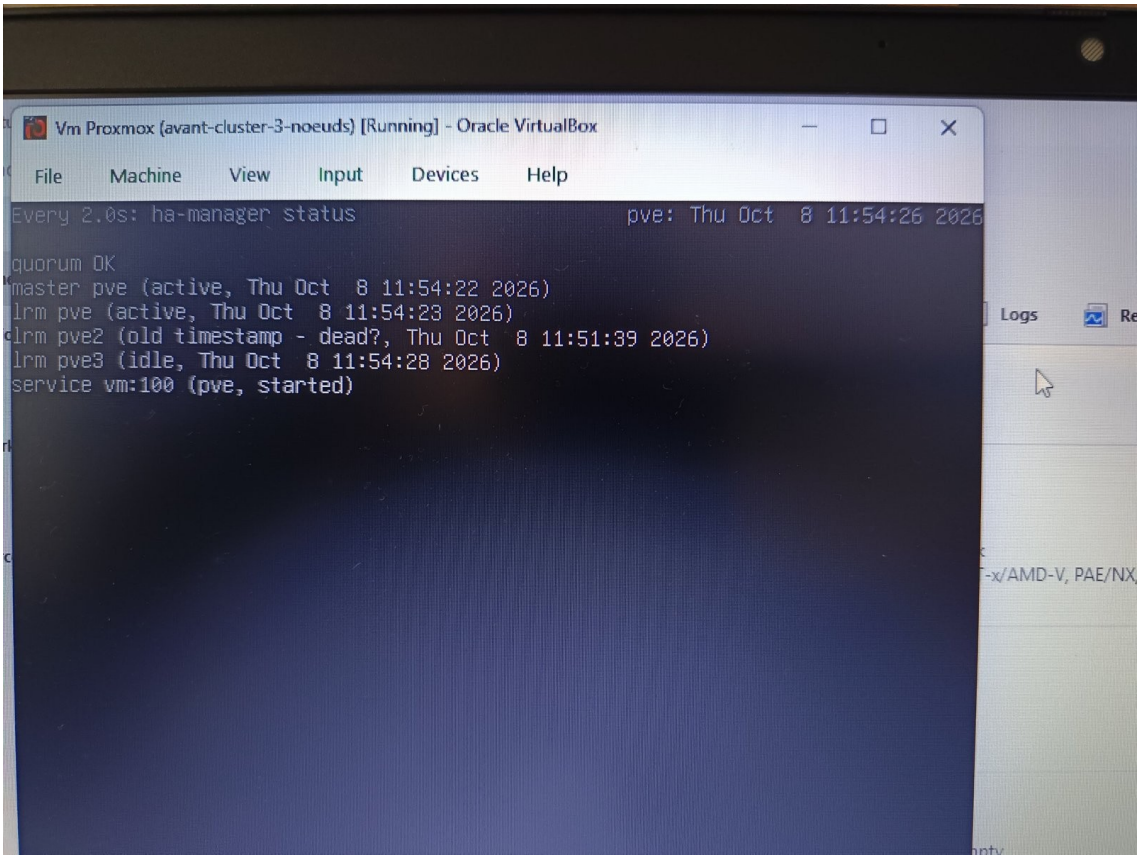
Monitoring mechanism on the host node (pve2): watchdog-mux is active and uses the "Software Watchdog"; the softdog module is loaded. If a node freezes, the watchdog reboots it, so it cannot keep running the VM while the VM is restarted elsewhere.



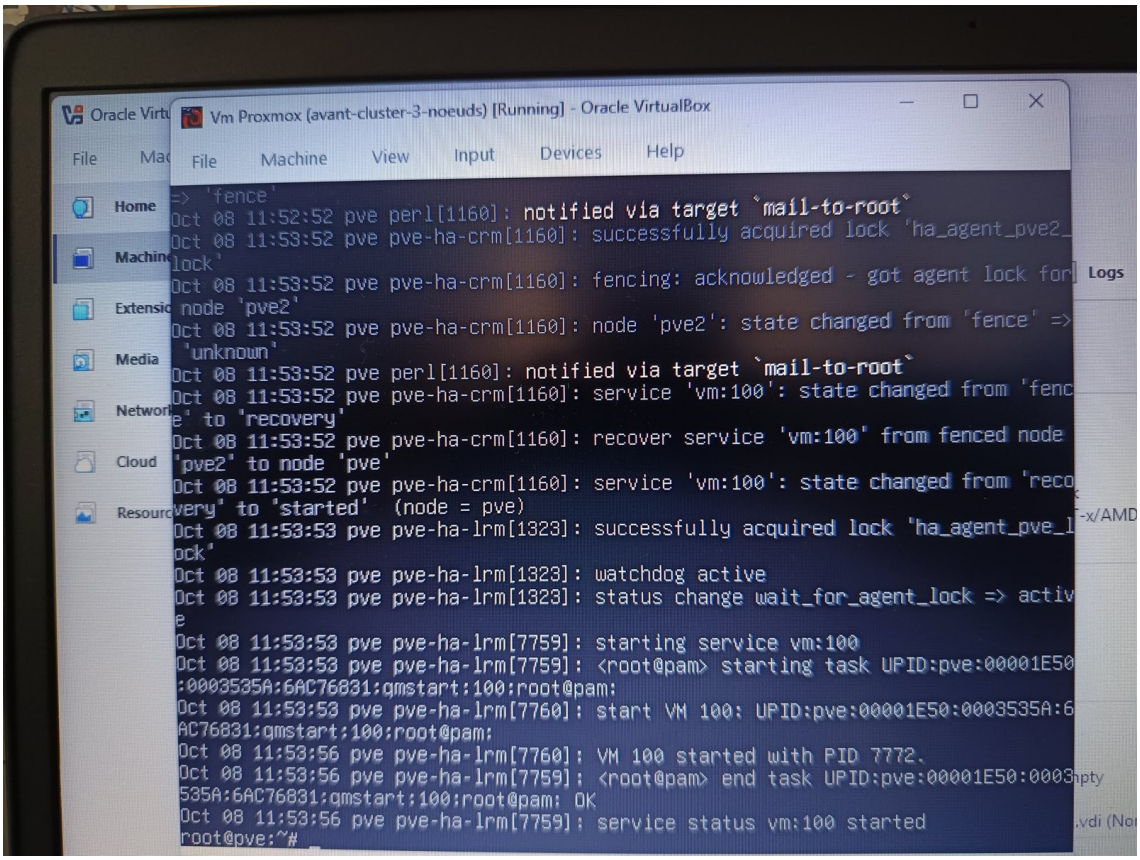
pve2: watchdog-mux active (running), Software Watchdog, lsmod | grep softdog

4.2 — Failure test and RTO measurement

Hard power-off of pve2 in VirtualBox (Machine > Close > Power off the machine). Times were read from the HA logs (journalctl -u pve-ha-crm -u pve-ha-lrm).



ha-manager status after the failure: pve2 "old timestamp - dead?", vm:100 (pve, started)



HA logs: fencing of pve2, recover service vm:100 to pve, VM 100 started

Measurement	Value
t0: node stopped (last signal from pve2)	11:51:39
t1: service restarted (ha-manager)	11:53:52
t2: VM usable again (VM 100 started)	11:53:56
Measured RTO (t2 minus t0)	2 min 17 s
Target RTO in our requirements spec	Not set in the spec ("to be confirmed with the client")
Gap	Cannot be calculated; for comparison, the March outage at Évry lasted over 6 hours

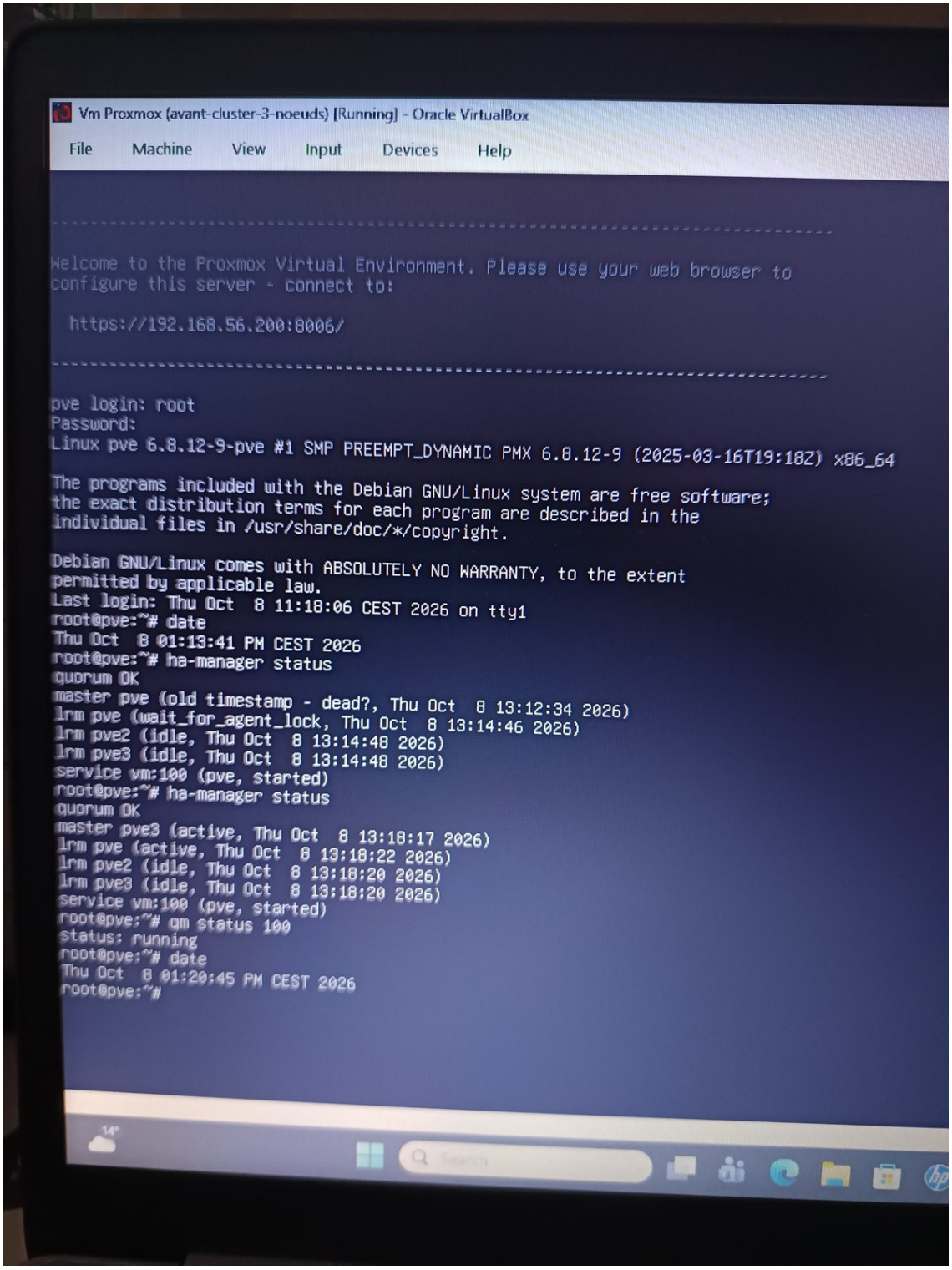
Analysis: almost all of the delay (about 2 min 13 s) is the safety wait during which Proxmox makes sure pve2 is really stopped (fencing) before restarting the VM. The restart itself took 4 seconds. HA chose to restart the VM on pve (not on pve3): HA picks the target node among the available nodes.

Limits: t0 is pve2's last signal, a few seconds before the actual power-off. VM-Test has no operating system: "usable" here means "VM started". With a real system, its boot time would have to be added (roughly 30 to 60 s).

Restoring normal state

pve2 was powered back on and rejoined the cluster (Nodes: 3, Quorate: Yes). The VM stayed on pve: Proxmox does not automatically move a VM back to its original node (no "failback"), to avoid a second interruption.

Incident: after the host PC went to sleep, the clocks of pve and pve3 were about 36 minutes behind (HA reported "detected time drift"). The sudden time correction (chronyc makestep) caused pve to be rebooted automatically by its watchdog (HA self-protection). HA then put everything back in order on its own: the master role moved to pve3 and VM-Test was running again. Lesson: time synchronisation (NTP) is essential in a cluster.



Final state: quorum OK, master pve3, vm:100 (pve, started), qm status 100: running

Comprehension questions

1. Why can't the VM restart elsewhere if its disk is stored only on the failed node?

A VM's disk is a file (or volume) holding its whole system and data. If it is stored on the failed node's local disk, the other nodes cannot access it: they can read the VM's configuration, but not its disk. Without a disk, the VM cannot start. That is why we put the disk on shared storage (NFS), accessible by all 3 nodes. Another solution is ZFS replication, which regularly copies the disk to the other nodes.

2. What is the difference between a BCP (PCA) and a DRP (PRA) in the Santélia case?

The BCP (Business Continuity Plan, PCA) keeps the service running during a failure, with no interruption or a very short one. For Santélia: if a server fails, the cluster automatically restarts the appointment or patient-records software on another server within minutes (2 min 17 s in our test).

The DRP (Disaster Recovery Plan, PRA) restarts the service after a more serious disaster, when the whole site or the data is affected: fire in the server room, ransomware. Data is restored from backups (Proxmox Backup Server, off-site copy, 3-2-1 rule). Recovery takes longer (several hours) and data since the last backup may be lost (RPO).

3. Does your measured RTO meet your requirements spec? If not, what should change: the technical solution or the requirement?

Our requirements spec does not yet set a numeric RTO: it is marked "to be confirmed with the client". So we cannot conclude with certainty. Our measured RTO of 2 min 17 s is far below the March outage (over 6 hours). If management approved, for example, an RTO of 15 minutes for critical applications, the solution would meet it. If they required less than one minute of downtime, the technical solution would have to change (active-active applications, load balancing), which costs more.

Improvement point: in our lab, the NFS share is hosted on pve. If pve fails, the shared storage disappears: it is a single point of failure. In production, the shared storage itself would need to be redundant (NAS in high availability, or Ceph), and the client should approve numeric RTO and RPO targets.